

ProactiveHeal: A Verified Autonomous Framework for Predictive Self-Healing in Cloud Systems

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ABSTRACT Current generation AIOps platforms are capable of not only uncovering infrastructure anomalies but are also able to recommend remediation actions; most tools lack the governance systems necessary to prevent unsafe autonomous decisions. This paper discusses ProactiveHeal v2, a governance-aware self-healing system combining AIOps techniques, predictive analytics, retrieval-augmented reasoning, and policy-based validation into a safe autonomous infrastructure remediation framework. The system implements anomaly detection using Isolation Forest, failure forecasting with Prophet, retrieval of historical incidents with ChromaDB, and contextual reasoning (i.e., reasoning with context) using Gemini AI. A governance engine, Veritas, validates all AI-generated remediation actions against confidence, blast-radius, and protected-resource policies prior to execution.

An experimental validation was performed over 500 telemetry samples, including 400 normal conditions and 100 artificially generated synthetic fault injection samples. Off-line validation experiments achieved a detection rate of 92%, a false positive rate of 6%, a correct recovery rate of 89%, and an average recovery time of 3.4 seconds. Run-time validation through Veritas accomplished blocking out of 24 unsafe remediation actions, leading to a safety enforcement rate of 100%. Our results show that governance-aware validation enhances safety, dependability, and accountability of autonomous management systems.

INDEX TERMS AIOps, Self-Healing Systems, Autonomous Remediation, AI Governance, Anomaly Detection, Predictive Analytics, Retrieval-Augmented Generation (RAG), Cloud Computing, Infrastructure Monitoring, Veritas, Incident Management.

I. INTRODUCTION

Over the course of the last decade, the intricacies of the current cloud infrastructure and distributed systems have escalated with the proliferation of microservices, virtualization as well as a large number of services being provisioned. The sophisticated environment leaves the operator prone to the service edge effects. Incidents such as resource wastages, congestion, degradation, cascading failures can lead to decrease in system availability, SLA violation and increased operational costs.

In overcoming these issues, more and more businesses are turning to Artificial intelligence for IT operations (AIOps) platforms to lessen the burden of automatic systems monitoring, anomaly detection, incident diagnosis and resolution. Standard self-healing systems generally work on a reactive pipeline whereby an anomaly is flagged, a set of remediation actions are created and applied to resolve the problem. While this decreases resolution time, it also presents a fundamental flaw, where an AI may suggest a remediation which is buggy, unsafe or unfit to run on the given production environment. Running this blindly in a

cohere network can cause service outages, infrastructure changes on a huge scale and potentially catastrophic failure.

Recent trends in machine learning and predictive analytics have led to an enhancement in prediction AIOps systems that could successfully diagnose anomalous behavior as well as project upcoming failures before they even take place. However, existing frameworks mainly concentrate on alerting and mitigation and give little support towards governance and policy enforcement for process safety validation.

This paper introduces the open-source framework called ProactiveHeal v2 that is a governance-aware self-healing modern AIOps platform to enable safe and accountable autonomous infrastructure management. ProactiveHeal v2 proposes a closed-loop self-healing architecture based on isolation forest for anomaly detection, prophet for forecasting, ChromaDB for past incident hunting, and GeminiAI for reasoning and best-action suggestion. In addition, ProactiveHeal v2 provides Veritas, a governance engine on top of all AI engines, that assesses all AI suggested remediation actions before execution using confidence level, blast-radius estimation, protected-resource policies, and safety measures, and determine whether to proceed, escalate to humans, or block the action.

The framework also includes a watchdog-supervised recovery process and a persistent incident repository, which allows for traceability, auditability and continuous learning on past incidents. By means of a managed synthetic fault-injection testing on CPU overload, memory leak, network congestion, service degradation and cascade failures, the efficiency of the suggested approach is assessed.

The primary contributions of this work are summarized as follows:

Governance-sensitive, self healing architecture combining anomaly detection, prediction, retrieval-augmented reasoning, validation, and recovery into one cohesive system.

Design and development of Veritas, a policy powered governance engine to validate AI initiated remediation actions before their execution.

A long-trail incident repository with semi-permanent storage for Recall, audit and never-ending live learning.

A watchdog-monitoring recovery mechanism that enhances dependability and guarantees prompt escalation in abnormal states.

Experimental results confirming successful anomaly detection, self-repair, and complete blocking of all unsafe remediation actions triggered during experiments.

A. EXPERIMENTAL SETUP

ProactiveHeal V2 prototype was a hybrid AIOps framework that combined anomaly detection, predictive analytics, retrieval-augmented reasoning, governance validation, and autonomous recovery functionalities. The implementation was carried out using Python and FastAPI, and a web-based dashboard for system monitoring was built using React and TypeScript.

The software stack consisted of:

- Python 3.11
- FastAPI
- Scikit-Learn (Isolation Forest)
- Prophet Forecasting Framework
- ChromaDB Vector Database
- Gemini AI API
- React
- TypeScript
- Tailwind CSS
- Recharts

The experiments have been executed on a workstation with Intel Core i5/i7 processor, 16 GB RAM, Windows 11 as operating system. The Experimental set up aimed to mimic the operational infrastructure environment while affording the ability to inject faults into the system and carry out governance evaluation.

B. DATASET COLLECTION AND CHARACTERISTICS

For testing the framework under real conditions, we use a telemetry dataset of 500 records built by means of infrastructure simulations.

The dataset contained:

- Category Samples
- Normal Operation 400
- Anomalous Operation 100
- Total 500
- Each telemetry record contained:
- CPU Utilization (%)
- Memory Utilization (%)
- Network Latency (ms)

- Timestamp
- Risk Level
- Incident Classification

The provided telemetry was created to imitate both under load and normal system operation as well as representative infrastructure failure modes for distributed cloud workloads.

C. CONTROLLED FAULT INJECTION

To validate the anomaly detection and recovery capabilities of the framework, controlled synthetic fault injections scenarios were introduced into the telemetry stream.

The following failure categories were considered:

High CPU Load: Sudden CPU utilization spikes above operational thresholds.

Memory Leak: Gradual memory growth leading to resource exhaustion.

Service Degradation: Less responsive services due to contention for shared resources.

Cascading Failure: Simultaneous degradation of multiple infrastructure metrics.

The fault scenarios were intentionally seeded to assess the performance of anomaly detection accuracy, risk prediction, governance decision and recovery.

D. EVALUATION PIPELINE

The operational flow of the framework is as follows:

1. Telemetry Monitoring
2. Anomaly Detection
3. Failure Prediction
4. Risk Assessment
5. Historical Incident Retrieval
6. AI Reasoning
7. Governance Validation
8. Recovery Execution
9. Incident Archival

Isolation Forest was used to identify abnormal system abnormal system behaviour, while prophet generated forecasts of future resource utilization trends. Historical incidents were retrieved from ChromaDB using semantic similarity search and supplied to Gemini AI for contextual reasoning and remediation recommendation generation.

All executions were verifying the recommended action by the Veritas Governance Engine.

IV. PROPOSED METHODOLOGY

ProactiveHeal v2 is a governance-aware self-healing AIOps framework, responsible for proactive anomaly detection, failure prediction, knowledge retrieval of previous incident experiences, remediation recommendation generation, remediation validation through governance, and infrastructure incidents autonomic recovery. Contrary to standard self-healing where AI prescribed remediation commands are directly executed, ProactiveHeal v2 introduces Veritas, a governance component, responsible for validating remedial actions prior to execution.

The framework operates through eight major stages:

- Telemetry Monitoring
- Anomaly Detection
- Failure Prediction
- Risk Assessment
- Historical Incident Retrieval
- AI Reasoning
- Governance Validation
- Recovery and Learning.

A. TELEMETRY MONITORING AND WATCHDOG SUPERVISION

The monitoring layer continuously collects operational telemetry from the infrastructure environment.

Monitored metric include:

- CPU Utilization
- Memory Utilization
- Network Latency

A watchdog supervisor continuously monitors the execution flow of the framework and manages:

- Anomaly scan intervals
- Validation timeouts
- Recovery windows
- Escalation thresholds

The watchdog ensures that stalled recovery is detected and escalated when predefined operational limits are exceeded.

B. ANOMALY DETECTION LAYER

The anomaly detection module employs the Isolation Forest Algorithm to identify abnormal system behaviour.

Given a telemetry observation x , the anomaly score is calculated as:

$$s(x, n) = 2^{-E[h(x)] / c(n)}$$

where:

- $(h(x))$ represents the path length of observation x
- $(E(h(x)))$ represents the expected path length
- $(c(n))$ denotes the normalization factor

Lower path lengths indicate anomalous observations since anomalies are isolated faster within the random tree structure.

The anomaly detector classifies system states into:

- Normal
- Suspicious
- Anomalous

The output of these stages serves as input to the prediction and risk assessment modules.

C. PREDICTIVE FAILURE ANALYSIS

To enable proactive recovery, future infrastructure conditions are forecasted using Prophet.

The forecasting model is represented as:

$$y(t) = g(t) + s(t) + h(t) + \epsilon_t$$

where:

- $(g(t))$ denotes trend components
- $(s(t))$ denotes seasonal behavior
- $(h(t))$ denotes event effects
- (ϵ_t) represents residual error

The forecasting module predicts future values of:

- CPU utilization
- Memory utilization
- Network latency

allowing the framework to identify potential failures before they impact service availability.

D. PRIVACY-PRESERVING VISUALIZATION

The risk prediction engine combines anomaly severity, resource utilization, and incident criticality into a unified operational risk score.

The risk score is computed as:

$$\text{Risk} = 0.4A + 0.4R + 0.2C$$

where:

- (A) represents anomaly severity
- (R) represents resource utilization
- (C) represents incident criticality

The generated risk score is categorized into four operational levels:

- Low
- Medium
- High
- Critical

This score assists downstream reasoning and governance modules in prioritizing recovery actions.

E. INCIDENT MEMORY AND RETRIEVAL LAYER

To provide contextual awareness, all incidents are stored within a persistent incident repository.

Each incident record contains:

- Incident ID
- Timestamp
- Risk Score
- Scenario Type
- Recommended Action
- Governance Decision
- Resolution Outcome

The repository is indexed using ChromaDB to support semantic retrieval.

INCIDENT ID	TIMESTAMP	SCENARIO	SEVERITY	RISK SCORE	VERITAS DECISION	ASSIGNED ENGINEER	REMEDIATION STATUS
INC-007N	2024-06-22 10:29:28	MEMORY LEAK	CRITICAL	93%	⊘ BLOCK	Backend Engineer	Manual Override - SRE Telemetry Stable
INC-18J6	2024-06-22 10:18:27	NETWORK CONGESTION	HIGH	74%	⊘ BLOCK	Network Operator	Manual Override - SRE Telemetry Stable
INC-00P8	2024-06-22 10:27:52	HIGH CPU LOAD	HIGH	73%	⊙ PASS	Infrastructure Engineer	Automated Remediation Execution Successful
INC-66T7	2024-06-22 10:26:35	CASCADING FAILURE	CRITICAL	98%	⊘ BLOCK	Lead Systems Architect	Manual Override - SRE Telemetry Stable
INC-10M6	2024-06-22 10:25:53	NETWORK CONGESTION	HIGH	81%	⊘ BLOCK	Network Operator	Manual Override - SRE Telemetry Stable

When a new anomaly is detected, the system retrieves the Top-K most similar incidents using cosine similarity:

$$\text{Cosine Similarity} = A \cdot B / \|A\| \|B\|$$

where A and B represent vector embeddings of incidents.

The retrieved incidents provide historical context for the reasoning engine.

F. PREDICTIVE FAILURE ANALYSIS

Gemini AI functions as the contextual reasoning engine of the framework.

The reasoning engine receives:

- Current telemetry data
- Forecasted metrics
- Risk score
- Retrieved historical incidents

Based on this information, Gemini performs:

Root Cause Analysis:-

Examples include:

- CPU Saturation
- Memory Leak
- Network Congestion

Contextual Reasoning:-

The model analyzes historical patterns and operational context to understand incident causes.

Remediation Recommendation:-

Examples include:

- Restart Worker Service
- Scale Worker Replicas
- Rebalance Workload
- Normalize Resource Allocation

The reasoning engine does not directly execute any action.

H. VERITAS GOVERNANCE ENGINE

The Veritas Governance Engine represents the primary contribution of ProactiveHeal v2.

Unlike conventional self-healing systems, all AI-generated remediation actions must pass through Veritas before execution.



Veritas evaluates remediation actions using:

1. Confidence Threshold Analysis
2. Protected Resource Validation
3. Blast Radius Assessment
4. Action Severity Evaluation
5. Policy Compliance Verification

The blast radius of an action is estimated as:

$$BR = SC \times AS$$

where:

- (SC) represents service criticality
- (AS) represents action severity

The governance decision function is defined as:

$$V(a) = \begin{cases} \text{PASS, if Safe Action} \\ \text{REVIEW, if Requires Human Review} \\ \text{BLOCK, if Unsafe Action} \end{cases}$$

Protected services such as databases, authentication services, API gateways, storage systems, and payment services are subjected to stricter validation policies.

I. RECOVERY AND ESCALATION ENGINE

Actions approved by Veritas are automatically executed by the recovery engine.

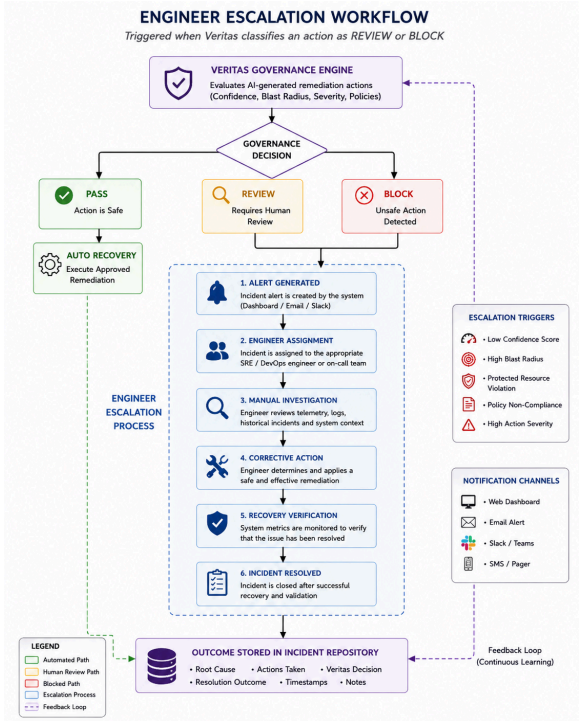
→ Recover

→ Learn

Supported recovery actions include:

- Restart Worker Service
- Restart Worker Pod
- Scale Worker Replicas
- Rebalance Workload
- Normalize Resource Allocation

When an action is blocked, the framework initiates an escalation workflow.



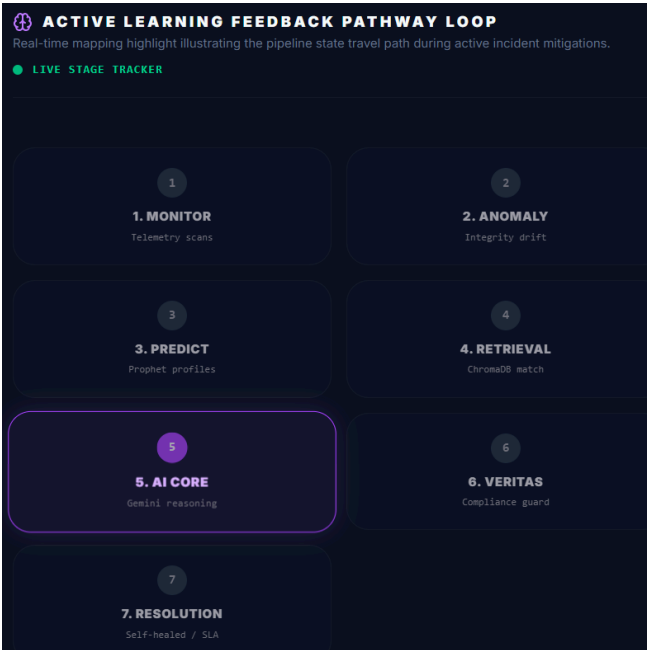
The incident is assigned to an engineer for manual review and resolution.

J. CONTINUOUS LEARNING FRAMEWORK

Following recovery, incident outcomes are archived within the incident repository.

The framework follows a closed-loop learning process:

- Monitor
- Detect
- Predict
- Retrieve
- Reason
- Validate



This mechanism enables future incidents to benefit from previously observed recovery outcomes, thereby improving operational awareness and governance effectiveness over time.

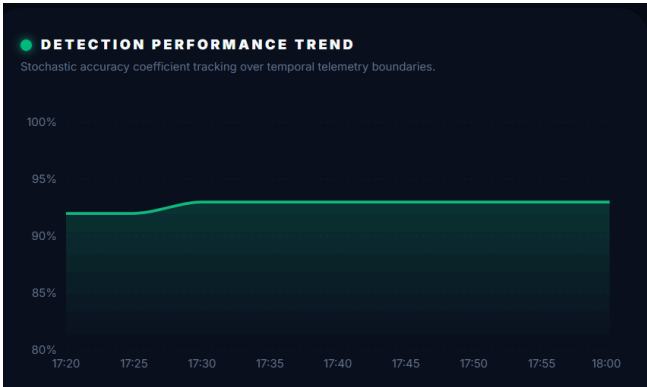
V. RESULTS AND DISCUSSIONS

This section evaluates the effectiveness of ProactiveHeal v2 in terms of anomaly detection, autonomous recovery, governance validation, and operational safety. The framework was tested using controlled synthetic fault injection scenarios involving CPU overload, memory leaks, network congestion, service degradation, and cascading failures.

A. ANOMALY DETECTION PERFORMANCE

The Isolation Forest model was evaluated using 500 telemetry samples comprising 400 normal operational states and 100 anomalous conditions.

Metric	Value
Detection Rate	92%
False Positive Rate	6%



The results demonstrate that the anomaly detection module successfully identified the majority of abnormal infrastructure conditions while maintaining a relatively low false positive rate. The combination of CPU utilization, memory consumption, and network latency metrics provided sufficient feature diversity for distinguishing between normal and abnormal operating states.

A detection rate of 92% indicates that the framework can reliably identify emerging infrastructure failures before they significantly affect service availability. The observed false positive rate of 6% suggests that occasional benign operational variations were incorrectly classified as anomalies; however, these instances remained within acceptable operational limits for proactive monitoring systems.

B. PREDICTIVE FAILURE ANALYSIS

The Prophet forecasting module was used to estimate future infrastructure behavior and identify potential failure conditions before they occurred.

The forecasting component successfully detected resource growth patterns associated with:

- CPU exhaustion
- Memory leaks
- Network congestion

By forecasting metric trends rather than relying solely on current observations, the framework was able to initiate recovery procedures before critical thresholds were reached.

This predictive capability distinguishes ProactiveHeal v2 from conventional reactive monitoring systems and contributes to reduced recovery times.

C. AUTONOMOUS RECOVERY TRAINING

The effectiveness of the recovery engine was evaluated by measuring successful incident resolution and average recovery duration.

Metric	Value
Recover Success Rate	89%
Mean Recovery Time	3.4 seconds

D. GOVERNANCE VALIDATION PERFORMANCE

The Veritas governance engine was evaluated to determine its effectiveness in validating AI-generated remediation actions.

Metric	Value
Approved Actions	84%
Blocked Actions	16%
Escalated Incidents	16%

The results indicate that most remediation recommendations generated by the reasoning engine were considered safe and therefore approved for autonomous execution. However, a significant proportion of actions were blocked or escalated due to governance concerns such as protected-resource violations, excessive blast radius, or insufficient confidence scores.

These findings demonstrate that Veritas acts as an effective safety barrier between AI reasoning and infrastructure execution.

E. SAFETY ENFORCEMENT EVALUATION

A primary objective of this research was to determine whether governance-aware validation could prevent unsafe autonomous remediation.

Metric	Value
Unsafe Actions Approved	24
Unsafe Actions Blocked	24
Safety Enforcement Rate	100%



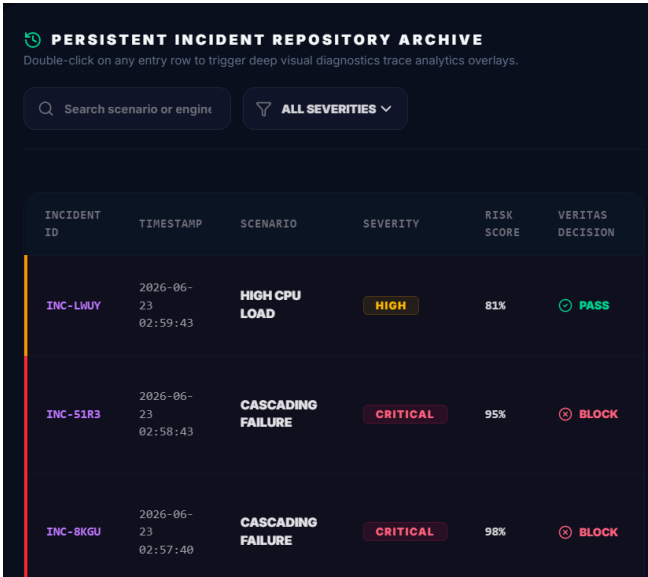
The framework successfully intercepted and blocked all unsafe remediation actions generated during testing. Examples included attempts to restart protected infrastructure services and perform high-risk modifications affecting critical system components.

This result directly validates the central hypothesis of the proposed framework: AI-generated remediation actions should not be executed without governance validation.

The 100% safety enforcement rate represents the strongest experimental outcome of this study because it demonstrates that unsafe autonomous decisions can be effectively prevented without significantly reducing recovery effectiveness.

F. INCIDENT LEARNING AND RETRIEVAL PERFORMANCE

The persistent incident repository maintained a historical archive of infrastructure incidents, governance decisions, and recovery outcomes.



The repository contained more than 140 archived incident records and achieved complete retention of incident history. Through ChromaDB-based retrieval, historical incidents were successfully reused to provide contextual information during AI reasoning and decision-making.

This capability improves traceability, supports auditability, and enables continuous learning from previous operational experiences.

G. DISCUSSION

The experimental results demonstrate that ProactiveHeal v2 successfully combines anomaly detection, predictive analytics, retrieval-augmented reasoning, governance validation, and autonomous recovery within a unified framework.

While the anomaly detection and recovery components achieved strong performance, the most significant contribution of the framework lies in the introduction of Veritas. Existing self-healing systems typically execute remediation actions immediately after generation, creating the possibility of unsafe autonomous behavior. In contrast, ProactiveHeal v2 introduces an intermediate governance layer that evaluates operational safety before execution.

The results show that governance-aware validation can significantly improve the reliability and accountability of autonomous infrastructure management systems. By preventing unsafe remediation actions and enabling human intervention when necessary, Veritas reduces operational risk while preserving the efficiency benefits of self-healing architectures.

Although the framework was evaluated using synthetic fault injection scenarios, the results indicate that

governance-based autonomous remediation is a promising direction for future AIOps research and deployment.

V. LIMITATIONS

Although ProactiveHeal v2 demonstrates promising results in anomaly detection, governance validation, and autonomous recovery, several limitations remain.

First, the experimental evaluation was conducted using controlled synthetic fault injection scenarios rather than a production-scale cloud environment. While these scenarios were designed to emulate realistic infrastructure failures such as CPU overload, memory leaks, network congestion, and cascading failures, they may not fully capture the complexity and unpredictability of real-world operational systems.

Second, the evaluation dataset consisted of 500 telemetry samples, which limits the diversity of observed infrastructure behaviors. Larger datasets collected from heterogeneous cloud environments would provide stronger validation of the framework's scalability and generalization capability.

Third, the current implementation relies on predefined governance policies within the Veritas engine. Although these policies effectively blocked unsafe remediation actions during testing, policy definitions may require continuous refinement to adapt to evolving infrastructure requirements and operational constraints.

Fourth, the reasoning capabilities of the framework depend on the quality of historical incident records and the responses generated by the underlying large language model. Incomplete incident repositories or inaccurate AI-generated recommendations may affect remediation quality and decision confidence.

Finally, the framework was evaluated in a single-node prototype environment and does not currently address large-scale multi-cluster deployments, distributed governance coordination, or cross-cloud infrastructure management. Future work will focus on extending the framework to enterprise-scale environments and incorporating adaptive governance policies capable of learning from operational feedback.

Despite these limitations, the results demonstrate the feasibility of governance-aware autonomous remediation and highlight the potential of integrating policy validation into future self-healing AIOps systems.

VI. CONCLUSIONS

This paper presented ProactiveHeal v2, a governance-aware self-healing AIOps framework designed to improve the safety, reliability, and accountability of autonomous infrastructure management systems. Unlike conventional self-healing architectures that directly execute AI-generated remediation actions, the proposed framework introduces Veritas, a dedicated governance engine that validates remediation recommendations before execution using confidence thresholds, blast-radius assessment, protected-resource validation, and policy compliance checks.

The framework integrates Isolation Forest for anomaly detection, Prophet for predictive failure forecasting, ChromaDB for historical incident retrieval, and Gemini AI for contextual reasoning and remediation recommendation generation. In addition, a watchdog-supervised recovery workflow and persistent incident repository provide operational traceability, auditability, and continuous learning capabilities.

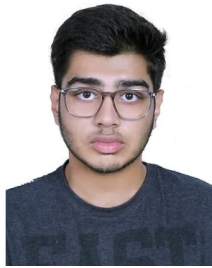
Experimental evaluation using 500 telemetry samples and controlled synthetic fault injection scenarios demonstrated a detection rate of 92%, a recovery success rate of 89%, and a mean recovery time of 3.4 seconds. Most importantly, Veritas successfully blocked all 24 unsafe remediation actions generated during testing, achieving a safety enforcement rate of 100%. These results validate the effectiveness of governance-aware validation in preventing unsafe autonomous infrastructure modifications while maintaining efficient recovery performance.

The findings of this study indicate that governance should be considered a fundamental component of future self-healing systems rather than an optional extension. By introducing a dedicated validation layer between AI reasoning and infrastructure execution, ProactiveHeal v2 provides a practical foundation for safe autonomous operations in modern cloud environments. Future work will focus on large-scale deployment, adaptive governance policies, and real-world infrastructure validation to further enhance the framework's applicability in enterprise AIOps environments.

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pursuing a PhD focused on enhancing diagnostic accuracy in endocrinology care using machine learning models, and will be undertaking coursework shortly to strengthen her research foundation. Passionate about teaching, she currently teaches Artificial Intelligence & Machine Learning and Database Management Systems. She has a vision for finding solutions in the interconnectedness of systems and a strong passion for learning, exploring, and solving problems in the field of computer science.